TECHNICAL SPECIFICATION & HARDWARE REPAIR MANUAL

SmartAttendance™ ESP32-P4 Biometric Terminal

Board-Level Schematics, Timing Constraints, Dual-Bank OTA Rollback Architecture, FreeRTOS Task Concurrency, and Exhaustive Hardware Troubleshooting & Repair Guide

- **SoC Core:** Espressif ESP32-P4 (Dual-Core RISC-V @ 400MHz, Hex-PSRAM @ 200MHz, AI Instructions)
- **Display Bridge:** 7.0″ 1024×600 MIPI-DSI (EK79007 bridge, 2-lane D-PHY @ 900Mbps, LVGL 9)
- **Vision Sensor:** OmniVision SC2336 2MP MIPI-CSI (24-pin FPC at top of screen, SCCB addr 0x36)
- **Touch Controller:** Goodix GT911 Capacitive Touch (Manual GPIO RST/INT sequence, 16-bit register IO)
- **Power & Charging:** Left slide switch (UP=ON), USB-C below switch (5V 2-3A), STC8 ADC telemetry
- **Fail-Safe Updates:** Dual-bank 6MB A/B flash partitions (`ota_0`/`ota_1`) with automated rollback
- **Storage Architecture:** SDMMC 1-bit FAT32 on bottom slot (`/sdcard`) with SQLite3 DB (DO NOT REMOVE!)
- **Board Repair:** Comprehensive PDN multimeter test points, component failure diagnosis & recovery

Target Audience:	Embedded Engineers, Firmware Devs, Hardware Techs	**Hardware Revision:**	CrowPanel Advanced 7″ ESP32-P4 v1.0
Framework / SDK:	ESP-IDF v5.3+ / ESP-DL v1.1.4+	**Typography:**	Times New Roman (12pt Body)
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[!] HARDWARE WARNING: Confidential Engineering Document

This manual contains low-level hardware schematics, register configurations, multimeter diagnostic test points, and component repair procedures intended exclusively for engineering and technical repair personnel. End-user operating guidelines are documented in the companion SmartAttendance User Manual.

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Section 1: Hardware Architecture & Electrical Pinout

The Elecrow CrowPanel Advanced 7″ ESP32-P4 v1.0 integrates the ESP32-P4 SoC with external Hex-PSRAM, MIPI CSI/DSI bridges, and power management circuits. Physical layout verified from hands-on testing:

- **Power Switch:** Physical slide switch located on the **left edge** (slide UP to connect battery/power, slide DOWN to disconnect).
- **Power / Charging Input:** USB Type-C port located on the **left edge directly below the slide switch**. Recommended power supply: **5.0V, 2.0A to 3.0A DC**.
- **Camera Module:** OmniVision SC2336 mounted along the **top edge of the screen** via 24-pin ribbon cable.
- **MicroSD Storage:** High-speed SDMMC slot located on the **bottom edge of the board**. **MANDATORY: DO NOT REMOVE DURING OPERATION!** The slot holds attendance.db and AI model weights.
- **Audio Subsystem Notice:** The audio amplifier and speaker have been permanently stripped from firmware v1.2 (2026-06-11). Audio defines in config.h are inactive tombstones.

Hardware Pinout & Bus Mapping Matrix

Peripheral	GPIO Pin	Interface / Net Name	Engineering Notes
Backlight PWM	GPIO 31	LEDC PWM Channel 0	30 kHz carrier frequency, 11-bit resolution (duty 200-2047).
Touch SDA	GPIO 45	I2C_NUM_0 SDA	Requires external 4.7kΩ pull-up to 3.3V.
Touch SCL	GPIO 46	I2C_NUM_0 SCL	100 kHz standard clock speed with pull-up.
Touch Reset	GPIO 40	GT911_RST	Must be driven via manual GPIO pulse to fix 0x5D address.
Touch Interrupt	GPIO 42	GT911_INT	Hold low during RST release to latch address 0x5D.
Camera CSI Data	DSI/CSI PHY	MIPI-CSI 2-Lane	OmniVision SC2336 24-pin FPC ribbon; RAW8 format at top of screen.
Camera SCCB SDA	GPIO 7	SCCB I2C SDA	Sensor address 0x36 (alternate address; NOT 0x30).
Camera SCCB SCL	GPIO 8	SCCB I2C SCL	Clock for camera sensor register read/write.
SDMMC Clock	GPIO 43	SD_CLK	SDMMC 1-bit clock line; 20 MHz to 40 MHz clock (bottom slot).
SDMMC Command	GPIO 44	SD_CMD	SDMMC command line with internal pull-up enabled.
SDMMC Data 0	GPIO 39	SD_D0	SDMMC data 0 bidirectional transfer.
Battery ADC	GPIO 4	ADC_CHANNEL_4	Monitors 3.3V to 4.2V battery via resistor divider network.
DS1302 RTC CLK	GPIO 51	J7 Header IO51	Serial clock for external battery-backed RTC.
DS1302 RTC DAT	GPIO 50	J7 Header IO50	Bidirectional serial data line for RTC register transfers.
DS1302 RTC CE	GPIO 49	J7 Header IO49	Chip Enable (active high) for DS1302 bus access.

[X] CRITICAL CONSTRAINT: Power Rail Sequencing (LDO3 & LDO4)

The EK79007 display bridge and MIPI PHY require internal LDO3 (2.5 V) and LDO4 (3.3 V) rails. If firmware fails to initialize these PMU regulators prior to display driver initialization, the MIPI-DSI link will fail to negotiate clock synchronization, leaving the panel permanently dark.

Section 2: MIPI-DSI Display & Backlight Subsystems

The display panel connects via a 2-lane MIPI-DSI interface managed by an EK79007 driver IC. Key physical parameters:

- **Lane Bit Rate:** 900 Mbps across 2 differential data lanes.
- **DPI Pixel Clock:** 51 MHz nominal.
- **Resolution & Porch Timing:** 1024×600 active pixels. HSYNC pulse: 20, HBP: 140, HFP: 160. VSYNC pulse: 3, VBP: 20, VFP: 12.
- **Critical Driver Constraint:** Do **NOT** call esp_lcd_panel_disp_on_off(). The EK79007 bridge does not support software display power toggle; attempting to do so locks the MIPI PHY.
- **LVGL 9 Optimization:** Allocate draw buffers strictly from PSRAM: swap_bytes=false, buff_spiram=true, sw_rotate=true, double_buffer=true.
- **Backlight Formula:** LEDC 30 kHz on GPIO31. Duty formula: duty = (pct != 0) ? ((pct * 18) + 200) : 0.

Section 3: GT911 Capacitive Touch Controller Quirks

The Goodix GT911 capacitive touch IC requires a precise hardware reset timing sequence to latch its I2C slave address at 0x5D. Standard generic drivers fail because they do not control RST and INT with sub-millisecond precision.

```
// 1. Install legacy I2C driver on I2C_NUM_0 (GPIO 45 SDA, GPIO 46 SCL, 100 kHz)
i2c_config_t i2c_conf = {
    .mode = I2C_MODE_MASTER, .sda_io_num = 45, .scl_io_num = 46,
    .sda_pullup_en = GPIO_PULLUP_ENABLE, .scl_pullup_en = GPIO_PULLUP_ENABLE,
    .master.clk_speed = 100000,
};
i2c_param_config(I2C_NUM_0, &i2c;_conf);
i2c_driver_install(I2C_NUM_0, I2C_MODE_MASTER, 0, 0, 0);

// 2. Hardware timing pulse to force address 0x5D:
gpio_set_direction(40, GPIO_MODE_OUTPUT); // RST
gpio_set_direction(42, GPIO_MODE_OUTPUT); // INT
gpio_set_level(40, 0); gpio_set_level(42, 0);  // Both driven low
vTaskDelay(pdMS_TO_TICKS(1));
gpio_set_level(40, 1);                          // Release RST high while INT stays low
vTaskDelay(pdMS_TO_TICKS(50));
gpio_set_level(42, 1);                          // Release INT high -> Latches 0x5D address!
vTaskDelay(pdMS_TO_TICKS(20));

// 3. Configure esp_lcd_panel_io_i2c (16-bit register commands are MANDATORY):
esp_lcd_panel_io_i2c_config_t io_cfg = {
    .dev_addr = 0x5D, .control_phase_bytes = 1, .dc_bit_offset = 0,
    .lcd_cmd_bits = 16,     // MUST be 16, NOT 8!
    .lcd_param_bits = 16,   // MUST be 16, NOT 8!
    .flags.disable_control_phase = 1,
};

// 4. Touch driver configuration (RST and INT set to NC because handled above):
esp_lcd_touch_config_t tp_cfg = {
    .x_max = 1024, .y_max = 600,
    .rst_gpio_num = GPIO_NUM_NC, .int_gpio_num = GPIO_NUM_NC,
    .flags = { .swap_xy = 0, .mirror_x = 1, .mirror_y = 1 },
};
esp_lcd_touch_new_i2c_gt911(tp_io, &tp;_cfg, &tp;);
vTaskDelay(pdMS_TO_TICKS(1000));
esp_lcd_touch_exit_sleep(tp); // Mandatory GT911 wake-up command
```

Section 4: MIPI-CSI Camera Driver & ISP Pipeline

The OmniVision SC2336 2MP image sensor is mounted at the top of the screen via a 24-pin FPC ribbon cable. Sensor parameters:

- **I2C Addressing:** SCCB slave address is strictly **0x36** (NOT 0x30).
- **Interface & Pixel Format:** 2-lane MIPI-CSI transmitting RAW8 Bayer BGGR at 24 MHz input clock.
- **ISP Pipeline:** The ESP32-P4 hardware ISP processes Bayer BGGR directly into RGB565 via DMA double-buffering.
- **Exposure Register Limits:** Total lines: 0x044C (1100 lines). Maximum safe integration time: reg 0x3E01 = 0x08, reg 0x3E02 = 0x90 (2192 lines max). Setting exposure higher causes sensor FIFO overflow and blank frames.
- **Camera Profiles:** Firmware provides 5 software presets adjusting digital gain and integration time: Auto, Outdoor (low gain/fast shutter), Indoor, Bright, and Low Light.

Section 5: High-Speed PSRAM & Flash Memory Architecture

The board integrates 16 MB or 32 MB Hex-mode PSRAM running at 200 MHz. Key configuration requirements in sdkconfig:

- **Hex-Mode PSRAM @ 200MHz:** CONFIG_SPIRAM_MODE_HEX=y, CONFIG_SPIRAM_SPEED_200M=y.
- **Experimental Symbol Suppression:** Must define #define ESP_IDF_VERSION_VAL(a,b,c) guards to prevent compilation failure when linking ESP-DL math kernels.
- **Heap Allocation Capabilities:** Display buffers and camera frame buffers MUST use MALLOC_CAP_SPIRAM | MALLOC_CAP_DMA. Internal SRAM (512 KB) is reserved strictly for FreeRTOS task stacks and critical ISR handlers.
- **Heap Sanity Checking:** Background task monitors free PSRAM every 30 seconds; logs warning if available PSRAM drops below 2.0 MB.

Section 6: Power Supply, Battery Telemetry & STC8 Gauge

The terminal incorporates a comprehensive power management and battery telemetry subsystem:

- **Slide Switch:** Physical mechanical switch on left upper side. Cuts battery and input power completely.
- **USB-C Power Input:** Located directly below slide switch on left edge. 5.0V, 2.0A–3.0A recommended.
- **STC8 Coprocessor:** An onboard STC8 microcontroller monitors battery current, voltage, and charger status via UART.
- **ADC Fallback Formula:** If STC8 is unresponsive, firmware reads GPIO4 ADC (1/2 divider): V_batt = (raw_adc / 4095.0) * 3.3 * 2.0 * 1.05.
- **Color States (No Audio / No Green):** Cyan Blue (#00CFFF) with lightning bolt when charging; White/Black normal (15%–100%); Red (#FF4444) when low (<15%).
- **Emergency Cutoff (5%):** Firmware detects critical threshold at 5% (3.40V) and initiates an automated atomic SQLite transaction close, flushes FAT32 caches, and powers down to prevent flash corruption.

Section 7 & 8: Deep Learning Vision Engine & Temporal Consensus

Face verification executes entirely on-device using a 2-stage neural inference pipeline accelerated by ESP32-P4 SIMD instructions:

- **Stage 1 Detection:** MobileNet-SSD detects faces and extracts 5 facial landmarks (eyes, nose, mouth corners) in ~22 ms.
- **Affine Landmark Alignment:** Normalizes facial roll, pitch, and yaw to standard 112×112 canvas.
- **Stage 2 Embeddings:** MobileFaceNet int8 quantized model outputs a 128-dimensional feature vector in ~38 ms.
- **Cosine Similarity Metric:** cos_sim = dot(u, v) / (||u|| * ||v||). Acceptance threshold: **0.68** (standard) / **0.72** (strict).
- **Multi-Frame Temporal Consensus:** Rather than triggering attendance on a single frame, the system evaluates a sliding window of 6 consecutive frames. Verification succeeds if **at least 4 out of 6 frames** match the same identity with confidence > 0.68. A 3-frame early exit fires if confidence exceeds 0.85.
- **Anti-Spoofing & Glare Filter:** Evaluates specular reflection histogram; rejects frames where saturated highlights cover > 18% of facial landmarks.

Section 9: FreeRTOS Task Architecture & Mutex Locking

The firmware executes across two symmetric RISC-V cores. Strict task priority and synchronization rules prevent race conditions:

Task Name	Priority	Stack Size	Core Affinity & Architectural Role
LVGL UI Task	10 (Highest)	8,192 bytes	Core 1. Handles screen rendering and touch event dispatch.
Detection Task	8	32,768 bytes	Core 0. Runs ESP-DL neural face detector and landmark alignment.
Camera Task	7	8,192 bytes	Core 0. Receives MIPI DMA interrupts and manages frame buffers.
Database Task	6	8,192 bytes	Core 1. Executes SQLite3 attendance writes and user queries.
Network Task	5	16,384 bytes	Core 0. Manages Wi-Fi, HTTPS sync API, and Telegram bot calls.
Battery Task	4	4,096 bytes	Core 1. Periodic 10s ADC voltage and STC8 telemetry polling.

LVGL Thread Safety Rule: Any non-display task invoking LVGL widget functions must acquire ui_acquire(). If ui_acquire() returns false (timeout), the caller MUST skip the UI call and bypass ui_release() to avoid corrupting the FreeRTOS recursive semaphore.

Section 10: Partition Table & Dual-Bank OTA Flashing Architecture

The 16 MB flash memory is partitioned for safe dual-bank A/B firmware upgrades with automated hardware rollback:

```
# Name,     Type, SubType, Offset,   Size,     Flags
nvs,        data, nvs,     0x9000,   0x6000,   # Non-volatile storage (PIN, Wi-Fi)
otadata,    data, ota,     0xf000,   0x2000,   # Dual 4KB sectors tracking active bank
phy_init,   data, phy,     0x11000,  0x1000,   # RF calibration parameters
ota_0,      app,  ota_0,   0x20000,  0x600000, # 6 MB Application Bank 0
ota_1,      app,  ota_1,   0x620000, 0x600000, # 6 MB Application Bank 1
storage,    data, spiffs,  0xc20000, 0x3d0000, # 3.8 MB SPIFFS assets & fallback HTML
```

Dual-Bank A/B State Machine & Rollback Protection

When an update is initiated via Web Portal (/update) or Telegram (/ota):
1. **Target Partition Selection:** esp_ota_get_next_update_partition(NULL) identifies the inactive bank.
2. **Streaming Flash Write:** Incoming firmware chunks are written via esp_ota_write() and validated with CRC32.
3. **Pending Verification State:** Upon restart, the bootloader flags the new partition as ESP_OTA_IMG_PENDING_VERIFY.
4. **Automated Self-Test & Validation:** The application initializes peripherals (PSRAM, SC2336 camera, SDMMC, GT911 touch). If all subsystems initialize successfully, main.c executes: esp_ota_mark_app_valid_cancel_rollback().
5. **Automated Rollback Trigger:** If a watchdog timeout (WDT), panic, or fatal exception occurs before this call, the bootloader automatically marks the new bank as ESP_OTA_IMG_INVALID and reboots into the previous working bank, ensuring zero device bricking.

Flashing Commands:
• **Full Clean & Build:** idf.py fullclean && idf.py build
• **Direct Flashing via COM port:** esptool.py --chip esp32p4 -p COM5 -b 460800 --before default_reset

```
--after hard_reset write_flash 0x0 build/bootloader/bootloader.bin 0x10000
build/partition_table/partition-table.bin 0x20000 build/SmartAttendance.bin
```

Section 11: SQLite3 Schema & On-Disk Data Structures

Biometric profiles and attendance events are stored in /sdcard/attendance.db on the bottom MicroSD slot. **MANDATORY: DO NOT REMOVE THE CARD!**

```
-- Users Table (Biometric Templates & Biographical Data)
CREATE TABLE IF NOT EXISTS users (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    uuid TEXT UNIQUE NOT NULL,
    name TEXT NOT NULL,
    student_id TEXT UNIQUE,
    phone_number TEXT,
    telegram_id TEXT,
    role TEXT NOT NULL CHECK(role IN ('student','lecturer','admin')),
    face_embedding BLOB,               -- 128-byte int8 MobileFaceNet vector
    created_at INTEGER NOT NULL,
    updated_at INTEGER,
    enroll_quality INTEGER DEFAULT 0,  -- Quality score (0 - 100)
    enroll_accepted INTEGER DEFAULT 0,
    enroll_rejected INTEGER DEFAULT 0,
    model_version INTEGER DEFAULT 1,
    embedding_dim INTEGER DEFAULT 128
);

-- Attendance Log Table
CREATE TABLE IF NOT EXISTS attendance (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER NOT NULL,
    session_id INTEGER,
    course_code TEXT,
    timestamp INTEGER NOT NULL,
    confidence REAL NOT NULL,
    synced INTEGER DEFAULT 0,          -- 0=local only, 1=synced to cloud
    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Security Credentials & Enrollment Security

• **Master Admin PIN:** Stored in NVS under key "admin_pin". Variable length: **4 to 8 numeric digits** (default initial PIN: 1234). Protects launcher apps, reports, settings, and manual enrollment authorization.
• **Dynamic Enrollment Session PIN:** Generated randomly in ble_registration.c via esp_random() % 1000000. Emitted as a **6-digit dynamic PIN** (format: %06d). Displayed in the screen banner and required for remote browser queueing over the Attendance_Setup Wi-Fi portal (which supports automatic captive portal detection).

Section 12: Board-Level Diagnostics & Hardware Repair Guide

Technicians diagnosing hardware failures, electrical anomalies, or communication breakdowns must adhere to this empirical repair protocol:

12.1 Power Distribution Network (PDN) Test Points & Voltage Rails

Use a calibrated digital multimeter (DMM) referenced to system ground (GND on J7 header) to verify power rails:

Power Rail / Net	Nominal Voltage	Acceptable Tolerance	Diagnostic Test Point / Pin	Failure Symptom if Out of Spec
VBUS_IN	5.00 V	4.75 V – 5.25 V	USB-C VBUS Pin / Input TVS Diode	If <4.6V, battery fails to charge; brownout resets during Wi-Fi transmit bursts.
VBAT	3.70 V (Nominal)	3.00 V – 4.20 V	Battery Terminal JST Header / Pin 1	If <3.0V, battery protection board trips; unit will not boot without USB-C connected.
VDD_3V3_SYS	3.30 V	3.20 V – 3.35 V	Main DC-DC Buck Inductor / J7 Pin 2	If <3.1V, ESP32-P4 SoC panics, QSPI flash bit flips, or SDMMC read timeouts occur.
LDO3_2V5 (PMU)	2.50 V	2.45 V – 2.55 V	PMU LDO3 Output Capacitor	Powers MIPI D-PHY. If disabled or <2.4V, D-PHY clock fails to lock; display remains black.
LDO4_3V3 (PMU)	3.30 V	3.25 V – 3.35 V	PMU LDO4 Output Capacitor	Powers EK79007 digital IO bridge. If disabled, panel displays no picture.
VBACKUP (RTC)	3.00 V	2.80 V – 3.30 V	DS1302 Coin Cell Holder / VCC Pin	If 0V, real-time clock loses time when power switch is DOWN; resets to 1970 on boot.

12.2 Component Failure Modes, Diagnostics & Field Repair Procedures

Case 1: MIPI-DSI Display Blank, White Screen, or Rolling Distortion

- **Root Causes:** 30-pin 0.5mm pitch FPC ribbon mis-seated; oxidation on gold contact fingers; LDO3 (2.5V) or LDO4 (3.3V) uninitialized; backlight boost circuit fault.
- **Diagnostic Procedure:** Connect oscilloscope to GPIO31 to verify 30 kHz PWM carrier. Check serial log for [esp_lcd_mipi_dsi] init errors. Measure LDO3 (2.5V) and LDO4 (3.3V).
- **Repair Steps:**
  1. Slide power switch DOWN and disconnect USB-C.
  2. Open 30-pin FPC flip-lock latch carefully using a plastic pry tool.
  3. Inspect ribbon gold fingers under 20x loupe for solder bridging, scratches, or oxidation. Clean contacts with 99.9% isopropyl alcohol (IPA) and lint-free swab.
  4. Reseat ribbon perfectly straight and press the lock down firmly until it clicks flush.
  5. If screen displays white with no pixels, replace the EK79007 bridge module.

Case 2: GT911 Capacitive Touch Controller I2C Arbitration Deadlock

- **Root Causes:** Missing 4.7kΩ pull-up resistors on GPIO45/46; cracked solder joints on 6-pin touch FPC header; I2C slave holding SDA low.
- **Diagnostic Procedure:** Probe GPIO45 (SDA) and GPIO46 (SCL) with logic analyzer. Idle level must be 3.3V. Inspect GPIO40 (RST) and GPIO42 (INT) timing on boot.
- **Repair Steps:**
  1. Verify that address latching sequence holds GPIO42 LOW while releasing GPIO40 HIGH.

2. **I2C Bus Deadlock Recovery:** If GT911 holds SDA low, toggle GPIO46 (SCL) 9 times manually to force the slave state machine to release the bus.
3. Re-flow solder on 6-pin 0.5mm touch header if cold solder joints are detected.

Case 3: SC2336 Camera Subsystem Probe Failure (SCCB Addr 0x36)

- **Root Causes:** 24-pin CSI ribbon inverted or unseated; camera lens out of focus; sensor address mismatch (must be 0x36, NOT 0x30).
- **Diagnostic Procedure:** Inspect serial log for esp_cam_sensor_detect: SC2336 not found. Probe 3.3V supply on camera FPC connector pins 1-2.
- **Repair Steps:**
  1. Re-latch the 24-pin ribbon at the top of the screen ensuring contacts face towards the PCB.
  2. Confirm firmware specifies .sccb_addr = 0x36.
  3. **Lens Focus Tuning:** If captured face images are blurry or yield low quality scores (<55), break the adhesive dot on the M12 lens thread and gently rotate clockwise/counter-clockwise until sharp focus is achieved at 60cm–90cm (2–3 feet) working distance.

Case 4: MicroSD Storage & SQLite Database Corrupt / Read-Only

- **Root Causes:** MicroSD card removed during active write; slot contact fatigue; dirty pads; FAT32 journal corruption.
- **Diagnostic Procedure:** Check serial console for sdmmc_req: sdmmc_send_cmd returned 0x107.
- **Repair Steps:**
  1. **Mandatory Rule:** MicroSD card MUST NOT be removed during operation.
  2. Inspect the bottom MicroSD socket pins. Clean card gold pads with IPA.
  3. If FAT32 filesystem is damaged, mount card on a PC and run fsck.vfat -a /dev/sdX1.
  4. To repair SQLite database, run: sqlite3 attendance.db "PRAGMA integrity_check;". If corrupted, restore from daily backup snapshot: cp attendance.db.bak attendance.db.
  5. Verify that /sdcard/models/p4/ contains all required MobileFaceNet neural weights.

Case 5: ESP32-P4 SoC Boot Loop, Brownout, or ROM Download Mode Lock

- **Root Causes:** Corrupted partition table or interrupted flash; strapping pin GPIO35 floating; brownout on 3.3V rail.
- **Diagnostic Procedure:** Connect USB-C to PC, open serial terminal at 115200 baud. Check ROM boot code.
- **Repair Steps:**
  1. Force UART download mode: Pull GPIO35 LOW while toggling EN pin.
  2. Erase entire flash: esptool.py --chip esp32p4 -p COMx erase_flash.
  3. Reflash complete partition suite: bootloader at 0x0, partition-table at 0x10000, and firmware at 0x20000.
  4. **OTA Boot Loop Recovery:** If the device is stuck in an OTA rollback loop, clear the otadata partition to restore factory boot default: esptool.py --chip esp32p4 -p COMx erase_region 0xf000 0x2000.

12.3 Diagnostic J7 Header Pinout & Test Points

The board exposes an unpopulated 10-pin header (J7) for logic analyzer probing and factory test automation:

J7 Pin	Signal Name	Type	Description & Diagnostic Use Case
Pin 1	GND	Power	System ground reference for oscilloscope and DMM probes.
Pin 2	VDD_3V3	Power	Main 3.3V digital supply rail test point.
Pin 3	GPIO 49	IO / RTC_CE	DS1302 Real-Time Clock Chip Enable line.
Pin 4	GPIO 50	IO / RTC_DAT	DS1302 Real-Time Clock Bidirectional Data line.
Pin 5	GPIO 51	IO / RTC_CLK	DS1302 Real-Time Clock Serial Clock line.
Pin 6	GPIO 35	Boot Strapping	Pull LOW during reset to force ESP32-P4 ROM UART Download mode.
Pin 7	TXD0 / IO37	UART Out	Primary debug UART transmit (115200 baud, 8N1).
Pin 8	RXD0 / IO38	UART In	Primary debug UART receive for factory test automation.

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